Toward a Tractability Frontier for Exact Relevance Certification

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April 6, 2026

Abstract

The optimizer-quotient manuscript archived at Zenodo [6] isolated six structurally nontrivial polynomial-time subcases for exact relevance certification, but the surrounding formal development now contains additional named tractable conditions, regime-specific lifts, and degenerate edge cases. We prove an explicit partition theorem for the current positive landscape. The fifteen currently formalized tractable families split into three disjoint blocks: six core structural subcases, four regime lifts, and five degenerate edge cases. Projecting that partition to primitive mechanisms yields a finite basis of size eight: the six core structural subcases from the earlier optimizer-quotient manuscript together with constant-optimizer collapse and finite explicit enumeration. Four stochastic/sequential tractable cases introduce no new primitive mechanism; they reduce directly to existing core subcases. The resulting decomposition isolates the currently verified positive structure and sharpens the open frontier question of whether exact relevance certification admits a finite optimizer-compatible tractability criterion analogous to a CSP-style algebraic boundary.

1 Introduction

The decision-quotient program now has a clean negative side and a clean positive side. The negative side was established in the optimizer-quotient manuscript archived at Zenodo [6]: exact relevance certification is regime-sensitive, hard in the general succinct setting, and constrained by the certification trilemma. The positive side began with six structurally meaningful polynomial-time subcases: bounded actions, separable utility, low tensor rank, tree structure, bounded treewidth, and coordinate symmetry.

Since then, the Lean development has grown beyond those six headline cases. It now contains regime-specific tractable lifts such as product distributions and bounded horizons, and also several edge cases such as single-action systems, bounded state spaces, strict dominance, constant optimal sets, and multiplicative-separable constant-sign models. This creates an immediate classification question. When we say the artifact contains "more tractable subcases," do we mean that the positive side is already proliferating without control, or do these extra cases reduce to a smaller conceptual basis?

We give an explicit finite decomposition of the currently formalized tractable families.

Contribution

The contribution is a finite-basis theorem for the current positive landscape:

1. an explicit partition theorem: the fifteen currently formalized families split into six core families, four regime lifts, and five degenerate edge cases;

2. the currently mechanized stochastic/sequential tractable lifts reduce to the old six and do not create new primitive families;
3. the degenerate cases collapse into two buckets, constant-optimizer collapse and finite explicit enumeration;
4. projecting the partition to primitive mechanisms yields a finite basis of size eight;
5. the resulting formalized positive landscape is explicit and finite.

The current formalized inventory consists of fifteen families. Whether only finitely many optimizer-compatible tractability mechanisms exist in general remains open. The main open question is whether newly formalized families represent genuinely new mechanisms or lifted or degenerate presentations of an already finite basis.

2 Finite Basis for the Current Artifact

We distinguish between *formalized tractable families* and *primitive tractability mechanisms*. A formalized family is any tractable condition with its own theorem statement or formal module in the current development. A primitive mechanism is a smaller conceptual source of tractability that may explain several families at once.

For the present artifact, the primitive list is explicit:

1. the six core structural subcases from the optimizer-quotient manuscript [6],
2. constant-optimizer collapse, and
3. finite explicit enumeration.

Thus the current primitive basis has size eight.

Theorem 2.1[1] (Explicit Partition of the Current Formalized Landscape). *The fifteen currently formalized tractable families admit an explicit disjoint partition into three blocks:*

1. *six* core *structural families,*
2. *four* lifted *families that reduce to existing core subcases, and*
3. *five* degenerate *families that become tractable by optimizer collapse or finite explicit enumeration.*

Moreover, every currently formalized family is classified into exactly one of these three blocks by an explicit formal classification map.

Proposition 2.2[2] (Explicit Inventory). *The fifteen currently formalized families are exactly the following:*

1. *core: bounded actions, separable utility, low tensor rank, tree structure, bounded treewidth, coordinate symmetry;*
2. *lifts: product distribution, bounded support, bounded horizon, full observability;*
3. *degenerate: single action, bounded state space, strict global dominance, constant optimal set, multiplicative-separable constant-sign.*

[1] Lean: `FR1`, `FR2` [2] : `FR25`, `FR26-28`

Corollary 2.3[3] (Finite Basis for the Current Mechanized Landscape). *Projecting the partition of Theorem 2.1 to primitive mechanisms yields an explicit finite list of eight primitive tractability mechanisms. The list consists of the six core structural subcases*

bounded actions, separable utility, low tensor rank, tree structure, bounded treewidth, coordinate symmetry,

together with two degenerate mechanisms:

constant-optimizer collapse and *finite explicit enumeration.*

Proposition 2.4[4] (Explicit Primitive Basis). *The current primitive basis consists exactly of bounded actions, separable utility, low tensor rank, tree structure, bounded treewidth, coordinate symmetry, constant-optimizer collapse, and finite explicit enumeration.*

Proposition 2.5[5] (Partition Counts). *The three blocks from Theorem 2.1 have sizes*

$$6 \quad \textit{(core)}, \qquad 4 \quad \textit{(lifted)}, \qquad 5 \quad \textit{(degenerate)}.$$

The current formalized landscape therefore contains fifteen tractable families accounted for by eight primitive mechanisms.

Proposition 2.6[6] (Five Core Mechanisms Are Already Nondegenerate). *The current formal development already contains witness families showing that five core mechanisms are individually indispensable in the sense of basis coverage:*

1. *a low-rank-only witness family,*
2. *a separable-only witness family,*
3. *a bounded-actions-only witness family, and*
4. *a bounded-treewidth-only witness family, and*
5. *a symmetry-only witness family.*

Each of these families remains tractable by its designated mechanism while failing the currently formalized versions of the other independent core mechanisms.

Proposition 2.6 now settles the independent part of the basis question on the stable formal surface. Every core mechanism that is not merely contained in another is formally known to be genuinely needed.

More concretely, the current stable formal surface already contains five distinct kinds of "only-by" witness:

1. low-rank only,
2. separable only,
3. bounded-actions only, and
4. bounded-treewidth only, and
5. symmetry only.

[3] : FR3 [4] : FR29 [5] : FR2 [6] : FR101, FR105, FR106, FR108, FR112, FR113

Proposition 2.7[7] (The Legacy `TreeStructured` Predicate Is Too Weak). *There exists a dependency presentation satisfying the older `TreeStructured` predicate whose dependency graph has real treewidth greater than one. In particular, width-one containment requires the stricter parent-tree condition from Proposition 2.9, not merely the legacy monotone-order predicate.*

Proposition 2.8[8] (Cyclic Dependencies Recover Hardness). *The reduction family from the optimizer-quotient article already yields coNP-hard instances with cyclic dependency structure. Thus the tree-structured regime is not only sufficient for tractability; leaving it also reaches the hard side.*

Proposition 2.9[9] (Parent-Tree Structure Gives Width-One Decompositions). *Suppose a dependency presentation is tree-structured in the strict parent sense: every dependency points to a smaller coordinate and every coordinate has at most one parent. Then the induced dependency graph admits a genuine tree decomposition of width at most one.*

Proposition 2.9 is the mathematically honest containment theorem behind the intended "tree structure is the treewidth-one special case" slogan. The formal development distinguishes this parent-tree notion from the weaker monotone-order predicate currently named `TreeStructured` in the older tractability file.

Proposition 2.10[10] (Current Positive Interfaces Factor Through Role Classification). *The current positive theorem surface factors through the role classifier of Theorem 2.1. Concretely, the family-indexed interfaces for*

1. *subcase-facing tractability,*
2. *degenerate-case witnesses, and*
3. *complexity-class summaries*

all factor through the map

$$family \longmapsto role \in \{core, lifted, degenerate\}.$$

Proposition 2.10 is the formal reason the current landscape feels smaller than its file count. The exported positive API is already controlled by the role partition rather than by fifteen unrelated theorem schemas.

3 Lifts and Degeneracies

The extra named families beyond the original six split naturally into two groups.

Regime Lifts

Four stochastic/sequential positive results do not introduce new primitive mechanisms. They reduce directly to existing core subcases from the optimizer-quotient manuscript [6]:

1. product distributions reduce to separable utility;
2. bounded support reduces to bounded actions;
3. bounded horizons reduce to bounded treewidth;
4. full observability reduces to tree structure.

[7] : `FR114` [8] : `FR115` [9] : `FR100`, `FR110`, `FR111` [10] : `FR4`

These are mathematically useful because they transfer the static tractable theory into richer regimes. But they do not enlarge the primitive basis. They are best understood as *lifts*, not new frontier points.

Proposition 3.1[11] (Lifted Cases Introduce No New Primitive). *Each currently mechanized stochastic/sequential tractable case is classified into one of the six core structural subcases. In particular, the current stochastic/sequential positive results do not enlarge the primitive basis from Corollary 2.3.*

Degenerate Cases

The remaining currently mechanized positive cases are degenerate in a precise sense.

Single-action systems, strict global dominance, constant optimal-set systems, and multiplicative-separable constant-sign models all collapse the optimizer so strongly that the relevant-coordinate question disappears: the optimal-action set is constant across states, so the empty coordinate set is sufficient. These are not new structural tractability mechanisms for the relevance problem; they are constant-optimizer collapses.

Bounded state space behaves differently. It does not force a constant optimizer. Instead, it makes exact certification feasible by brute-force enumeration over a finite universe. This is polynomial only because the instance family already carries a hard external bound on its search space.

Proposition 3.2[12] (Degenerate Positive Cases). *Every currently mechanized non-core tractable case that is not a regime lift belongs to one of two degenerate buckets:*

1. *constant-optimizer collapse, or*
2. *finite explicit enumeration.*

Taken together with Proposition 3.1, Proposition 3.2 sharpens Theorem 2.1: every currently formalized positive result is exactly one of three things, a primitive core case, a lift, or a degenerate collapse.

4 Toward the Frontier Theorem

Three conjectures frame the frontier program.

Conjecture 4.1 (Finite Optimizer-Compatible Criterion). There exists a finite list of optimizer-compatible structural criteria such that every representation-level tractable class of exact relevance-certification instances satisfying the necessary conditions of Section 6 satisfies at least one criterion on that list.

Conjecture 4.2 (Finite Primitive Basis Beyond the Current Artifact). Every polynomial-time family for exact relevance certification arises either

1. from a primitive optimizer-compatible structural mechanism,
2. as a lift/reduction to such a mechanism, or
3. from a degenerate collapse analogous to constant-optimizer collapse or finite explicit enumeration,

and only finitely many primitive mechanisms are needed.

[11] : FR5 [12] : FR1, FR6

Conjecture 4.3 (Optimizer-Compatible Dichotomy). For representation-level structural classes of coordinate-structured decision problems satisfying the necessary conditions of Section 6, exact relevance certification is either polynomial-time by membership in one of the finitely many optimizer-compatible primitive mechanisms of Conjecture 4.2, or coNP-hard.

The three conjectures form a ladder. Conjecture 4.1 asks only for a finite conceptual description. Conjecture 4.2 adds the stronger claim that all tractable positive results factor into primitive, lifted, or degenerate forms. Conjecture 4.3 is the full boundary statement.

Three technical questions sit between the present paper and a true metatheorem.

First: Realizability

Theorem 5.1 shows that naive realizability is not the bottleneck. In the unconstrained setting, arbitrary labeling kernels already arise as optimizer quotients. The frontier theorem therefore cannot come from quotient realizability alone. It has to characterize optimizer-compatible structure at the level of representation-level structural classes, not merely at the level of abstract set partitions. Section 6 isolates the first necessary conditions any such statement must satisfy.

Second: Primitive Versus Derived Tractability

A genuine frontier theorem must distinguish primitive mechanisms from lifts and degenerate cases, and show that this distinction is not an accident of the current examples.

Third: Algebraic Description

The natural long-term analogy is the CSP dichotomy program, where tractability is controlled by an algebraic boundary condition rather than by an ad hoc list of examples. For exact relevance certification, the unresolved question is whether optimizer-induced quotients admit a comparably finite algebraic description or whether the tractable side remains genuinely open-ended.

5 The Realizability Barrier

A natural hope for a finite tractability taxonomy would be this: perhaps optimizer-induced quotients realize only a narrow subclass of equivalence relations, so the finite frontier would emerge from realizability alone. In the unconstrained setting, that hope is false.

Theorem 5.1[13] (Every Labeling Kernel Is Optimizer-Realizable). *Let $\phi : S \to T$ be any function. There exists a decision problem with action space T and state space S such that for every state $s \in S$,*

$$Opt(s) = \{\phi(s)\}.$$

Consequently, for all states $s, s' \in S$,

$$Opt(s) = Opt(s') \iff \phi(s) = \phi(s').$$

Equivalently, the decision quotient of the constructed problem is exactly the kernel partition of ϕ.

Proof. Take the action set to be T itself and define

$$U(a, s) = \begin{cases} 1 & \text{if } a = \phi(s), \\ 0 & \text{otherwise.} \end{cases}$$

[13] : FR7, FR8

Then the unique maximizer at state s is the designated action $\phi(s)$, so the optimizer set is exactly $\{\phi(s)\}$. Two states therefore have the same optimizer set if and only if they receive the same label under ϕ. ■

Theorem 5.1 is the key negative fact for the frontier program. It shows that optimizer realizability by itself is extremely permissive. If arbitrary label kernels are already optimizer-induced, then no finite taxonomy can come merely from asking which abstract quotients are realizable.

Proposition 5.2[14] (Every Equivalence Relation Is Optimizer-Realizable). *For every equivalence relation on the state space, there exists a decision problem whose decision quotient relation is exactly that equivalence relation. Moreover, the quotient object of the realizing problem is canonically equivalent to the corresponding setoid quotient.*

Proposition 5.2 is the maximal realizability statement available in the present framework. The labeling-kernel theorem is a special case. Thus the realizability barrier is not merely broad; at the level of abstract quotient relations, it is maximal.

Proposition 5.3[15] (The Realized Quotient Is Exactly the Label Range). *For every labeling map $\phi : S \to T$, the decision quotient of the realizing problem is canonically equivalent to the range of ϕ. In finite settings, the quotient cardinality is therefore exactly $|\operatorname{range}(\phi)|$.*

Proposition 5.3 strengthens Theorem 5.1. The realizing construction does not merely realize the induced equivalence relation abstractly; it realizes the quotient object itself with exactly the expected image size.

Corollary 5.4[16] (Realizability Alone Cannot Be the Frontier). *Any future optimizer-compatible tractability metatheorem must use more than quotient realizability alone. Additional restrictions have to enter through representation, coordinate structure, utility structure, closure properties of the family under study, or comparable algebraic invariants.*

This shifts the burden of the metatheorem. The hard question is not which quotients can arise in principle; almost all kernels can. The hard question is which *natural structural families of decision problems* force tractability or preserve hardness once coordinate structure and utility representation are taken seriously.

6 Necessary Conditions on Any Frontier Statement

The realizability barrier leaves a more precise frontier question. If quotient shape alone is too expressive, what should replace it as the organizing unit of the metatheorem?

At minimum, the tractability frontier should not classify arbitrary sets of decision problems. It should range over representation-level structural classes and it should ignore benign changes of presentation.

The right eventual condition list will almost certainly be stronger than this minimal principle. For example, a mature frontier theorem may also require closure under adding explicitly irrelevant coordinates, adjoining dummy actions, or other representation-preserving refinements. But action/state relabel invariance is the irreducible starting point.

Proposition 6.1[17] (Exact Certification Depends Only on the Decision Quotient Relation). *If two decision problems on the same state space induce the same decision-equivalence relation, then they induce the same exact-certification structure. In particular, they have the same sufficient coordinate sets, the same minimal sufficient coordinate sets, and the same relevant and irrelevant coordinates.*

[14] : FR45, FR46, FR47 [15] : FR39, FR40 [16] : FR7, FR8 [17] : FR30, FR31, FR32, FR33, FR34, FR35, FR37, FR51

Moreover, their quotient objects are canonically equivalent; in finite settings, those quotient objects therefore have the same cardinality.

Proposition 6.1 is the canonical statement behind the frontier program. Exact certification is determined by the decision quotient relation itself. Equality of optimizer maps is only a sufficient route to this conclusion, because equal optimizer maps induce the same decision-equivalence relation.

Corollary 6.2[18] (Sufficiency Is Relation Refinement). *A coordinate set I is sufficient if and only if the coordinate-agreement relation induced by I refines the decision quotient relation.*

Corollary 6.2 is the cleanest relational form of exact certification. It removes utility language entirely: sufficiency asks whether one state relation is contained in another.

Corollary 6.3[19] (Relevance Is Failure of Erased Sufficiency). *A coordinate i is irrelevant if and only if deleting it leaves a sufficient coordinate set, namely* $\mathrm{univ} \setminus \{i\}$. *Equivalently, i is relevant if and only if that erased set is not sufficient.*

Corollary 6.3 makes the role of individual coordinates canonical: relevance is exactly the obstruction to retaining sufficiency after coordinate erasure.

Corollary 6.4[20] (Certification Statistics Factor Through the Quotient Relation). *Any statistic that is a function of the sufficient-set family and relevant-coordinate set is invariant under equality of the decision quotient relation. In particular, the number of sufficient coordinate sets, the number of minimal sufficient coordinate sets, and the number of relevant coordinates are quotient invariants.*

Corollary 6.4 records the next logical consequence of Proposition 6.1. Once exact certification is identified with quotient data, every certification statistic built from sufficient sets or relevant coordinates factors automatically through that quotient data as well.

Corollary 6.5[21] (Constant-Optimizer Collapse Forces Trivial Certification). *If the optimizer set is constant across all states, then the empty coordinate set is sufficient and every coordinate is irrelevant.*

Corollary 6.5 isolates the first genuinely degenerate mechanism in the tractable basis. The certification problem disappears because there is no state dependence left to certify.

Proposition 6.6[22] (Explicit Enumeration Is Parameter-Dependent, Not Structural). *For every fixed exponent k, there exists a Boolean-cube family with state space size $2^n > n^k$. Thus explicit-state enumeration can outrun any fixed monomial bound in the ambient dimension parameter.*

Proposition 6.6 is weaker than a full polynomial lower-bound metatheorem, but it already captures the structural point needed here: tractability by brute-force state enumeration depends on a separate size parameter and does not arise from an optimizer-compatible mechanism like symmetry, low rank, or tree structure.

Proposition 6.7[23] (Positive Affine Utility Reparameterizations Are Invisible). *Statewise positive affine transformations of utility leave exact certification unchanged. In particular, replacing*

$$U(a,s) \qquad \textit{by} \qquad \alpha(s)+\beta(s)U(a,s)$$

with $\beta(s) > 0$ for every state s does not change sufficient coordinate sets or relevance judgments.

Proposition 6.7 is another direct consequence of Proposition 6.1. It records a standard utility-theoretic invariance in the exact-certification language: only the induced decision quotient matters, not the particular positive affine scale used to encode utility.

Proposition 6.8[24] (Duplicate Actions and Duplicate States Preserve Certification). *Duplicating a single action without changing its utility profile preserves the decision quotient relation and hence*

[18] : FR48 [19] : FR49, FR50 [20] : FR41, FR42, FR43, FR44 [21] : FR52, FR53 [22] : FR66, FR67 [23] : FR13-16 [24] : FR54, FR55, FR56, FR68, FR69, FR70, FR71, FR72, FR73

preserves sufficiency and relevance. At the optimizer-set level, the only possible change is the addition of the duplicate action itself: original optimal actions remain optimal, and the duplicate is optimal exactly when the original action was. Duplicating a single state without changing its utility profile preserves the decision quotient up to the obvious quotient equivalence and also preserves sufficiency and relevance.

Proposition 6.8 gives two more theorem-forced closure laws. These are not modeling conventions; they are consequences of the fact that exact certification factors through quotient data rather than through accidental multiplicity in the presentation.

Proposition 6.9[25] (Relabeling Invariance Is Forced by Exact Certification). *Action relabeling preserves optimizer equivalence and exact sufficiency. State relabeling does so as well once the coordinate structure is transported along the state bijection. Consequently, any structural tractability frontier for exact relevance certification must be closed under these relabelings.*

Proposition 6.9 is now best read as a corollary of Proposition 6.1. Bijective relabelings do not change the optimizer map up to the obvious identification, so they cannot change exact certification.

The second necessary ingredient is a genuine expressivity restriction. Call a class *kernel-universal* if, for every labeling map $\phi : S \to T$, it contains some decision problem whose optimizer quotient realizes exactly the kernel partition of ϕ. By Theorem 5.1, kernel universality is an extremely strong property.

The canonical finite invariant here is the size of the optimizer quotient, equivalently the number of distinct optimal-action sets. The optimizer-quotient manuscript [6] identifies the quotient with the range of the optimizer map, so quotient size is not auxiliary bookkeeping; it is intrinsic.

Proposition 6.10[26] (Quotient Size Is Unbounded Under Realizability). *For every $m \in \mathbb{N}$, there exists a finite decision problem whose optimizer quotient has size exactly m.*

Proof. Take $m = k+1$ and realize the identity labeling on $\mathrm{Fin}(k+1)$. Proposition 5.3 identifies the quotient object with the range of that identity labeling, which has cardinality $k+1$. Equivalently, Proposition 5.2 realizes the discrete equivalence relation on $\mathrm{Fin}(k+1)$. ■

The point is simple: a meaningful frontier statement cannot range over classes that are simultaneously relabel-invariant and kernel-universal, because quotient shape alone then becomes too expressive to isolate the boundary.

Proposition 6.11[27] (Invariance Alone Is Too Weak). *The universal class of all decision problems satisfies action relabeling invariance and state relabeling invariance, but it is kernel-universal. Consequently, relabeling invariance by itself is far too weak to isolate a tractability frontier.*

Proposition 6.11 separates two kinds of conditions that can otherwise be conflated.

1. **Benign invariance axioms** say the family should ignore arbitrary naming choices.
2. **Expressivity-limiting axioms** say the family should not realize arbitrary quotient structure wholesale.

Any genuine optimizer-compatible frontier theorem will need both. Without decision-quotient invariance and its immediate corollaries such as relabel invariance and positive affine invariance, the statement is not structural. Without an expressivity restriction excluding kernel universality or something comparably strong, the realizability theorem and Proposition 6.10 make the class too large for quotient shape alone to carry the boundary.

[25] : `FR17-20` [26] : `FR24`, `FR38` [27] : `FR21`, `FR22`, `FR23`

7 Related Work

Rough Sets and Reducts

At the static level, the closest classical neighbor remains rough-set reduct theory [4, 7]. That literature asks which attributes can be removed while preserving decision distinctions. The optimizer-quotient manuscript [6] showed that exact static sufficiency can be recast as reduct preservation for the induced decision table. The finite-basis theorem here instead classifies the currently formalized positive cases once the optimizer-quotient perspective is fixed.

Backdoors and Tractable Islands in CSPs

The nearest complexity-theoretic analogue is backdoor tractability in constraint satisfaction [8, 1, 3]. In that setting, a small structural set can expose membership in a tractable class even when the ambient problem is intractable. The parallel to the optimizer-quotient manuscript [6] is direct: exact relevance certification asks which coordinates matter, and structured tractability appears when the hidden source of hardness is restricted.

What this paper contributes relative to that literature is not a new backdoor algorithm. It is a classification layer over the currently formalized positive results. The finite-basis theorem and partition theorem show that the present formalized landscape already separates three phenomena that are easy to blur together: primitive tractable islands, regime-specific lifts into old islands, and degenerate cases where the optimizer collapses or the instance space is externally bounded.

Dichotomy Theorems

The long-term analogy is the dichotomy tradition for satisfiability and constraint satisfaction, from Schaefer's Boolean dichotomy theorem [5] to the finite-domain CSP dichotomy theorems of Bulatov and Zhuk [2, 9]. Those results do not merely enumerate examples; they characterize an entire tractability boundary. For exact relevance certification, Sections **??** and 6 isolate two constraints any analogous theorem would have to respect: the currently available positive cases already form a controlled family, while quotient realizability alone is too permissive to explain the frontier. Any optimizer-compatible analogue must therefore be representation-sensitive rather than quotient-existence-sensitive.

8 Conclusion

The finite-basis theorem and the realizability barrier isolate two constraints on any optimizer-compatible tractability frontier. First, the currently formalized positive cases already admit a structured decomposition into primitive mechanisms, lifts, and degenerate collapses. Second, quotient realizability alone is too permissive: arbitrary labeling kernels are already optimizer-realizable, so the frontier cannot be characterized at the level of quotient shape alone.

The remaining open problems are structural. One must determine whether the current primitive/lift/degenerate trichotomy extends beyond the presently formalized families, and one must identify expressivity-limiting conditions on representation-level classes in addition to the invariances forced by exact certification. Decision-quotient invariance, relabel invariance, and positive affine utility invariance are necessary, but they are not sufficient; the unbounded quotient-size construction shows that a successful frontier theorem must also rule out kernel-universal behavior.

9 Artifact Index and Supplementary Tables

This paper uses the same theorem-local proof-handle convention as the rest of the series: inline markers of the form `\LH{...}` and `\LHrng{...}` attach formal support directly to the corresponding paper claims.

The cited-handle closure for this manuscript comprises 9438 lines and 469 theorem-level declarations across 49 files; this is the code needed for the handles cited in the paper. The local `proofs_4d` development contains 4634 lines and 240 theorem-level declarations across 41 files, while the cumulative transitive dependency closure contains 76420 lines and 3331 theorem-level declarations across 304 files. Thus the new paper4d development contributes 6% of cumulative lines and 7% of cumulative theorem count. The supplementary handle ledger records source-file provenance for every cited handle: entries under `Paper4dFrontier/*` are new in this paper, whereas entries under `DecisionQuotient/*` are inherited from the earlier optimizer-quotient artifact [6].

The full handle ledger and claim-to-handle mapping are provided in the supplementary PDF and archived artifact. We omit those longer tables from the manuscript body.

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